Markov Chain

$$P(X_{n+1}=j \mid \underline{X_n=i}, \underline{X_{n-1}=i_{n-1}, \dots, X_0=i_0})$$

$$= P(X_{n+1}=j \mid \underbrace{X_n=i}) \quad \text{— Markov cond}$$

$$P = ((p_{ij}))$$

$$= P(X_1=j \mid X_0=i) \rightarrow \text{homogeneity}$$

$$= p_{ij} \rightarrow \text{transition prob}$$

Ex :- IId, RWs.

RW on graphs:

$G = (V, E)$, V = set of vertices.

E = set of edges between vertices.

$V = \{A, B, C, D, E\}$

$E = \{(A,B), (A,C), (B,C), (B,D), (B,E), (C,D), (D,E)\}$

For a vertex $v \in V$,

$\mathcal{N}(v) =$ nbrs of v.

$\deg(v) = |\mathcal{N}(v)|$

$=$ No. of elements in $\mathcal{N}(v)$.

$$p_{u,v} = \begin{cases} \frac{1}{\deg(u)} & \text{if } v \in \mathcal{N}(u) \\ 0 & \text{otherwise} \end{cases}$$

$\mathcal{N}(A) = \{B, C\}$

$\mathcal{N}(B) = \{A, C, D, E\}$

$\vdots$

$$P = \begin{array}{c|ccccc|} & A & B & C & D & E \\ A & & 1/2 & 1/2 & & \\ B & 1/4 & & 1/4 & 1/4 & 1/4 \\ C & 1/3 & 1/3 & & 1/3 & \\ D & & 1/3 & 1/3 & & 1/3 \\ E & & 1/2 & & 1/2 & \end{array}$$

<u>Extended version</u> : Assign weight $w_{uv} = w_{vu}$ for every pair of vertices u & $v \in V$.

Weights are non-neg., & some of them can be 0.

For any $u \in V$,

$$w_{u\cdot} = \sum_{v \in V} w_{uv} < \infty \quad \forall u \in V$$

$w_{AB} = 2$ | $w_{A\cdot} = 3$

$w_{AC} = 1$ | $w_{C\cdot} = 8$

$$p_{uv} = \frac{w_{uv}}{w_{u\cdot}}$$

($w_{uv} = 0$ if $(u,v) \notin E$.)

Note, the deg model can be obtained by setting $w_{uv} = 1/0$

Page rank modification

$\Gamma = \{E\}$.

Let Γ be a fixed subset of G.

Let $\alpha \in (0,1)$,

$V = \mathbb{Z}$

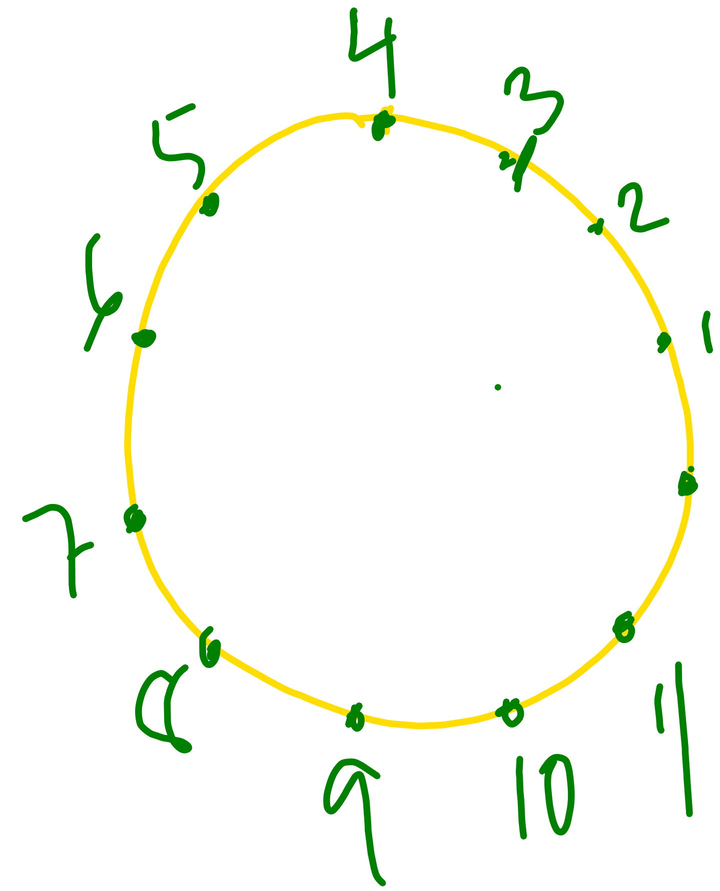
$E = \{(i,i+1) : i \in \mathbb{Z}\}$

$$p'_{uv} = (1-\alpha)\, p_{uv} + \alpha\, q_{uv}.$$

$$q_{uv} = \begin{cases} \frac{1}{|\Gamma|} & \text{if } v \in \Gamma \\ 0 & \text{o.w.} \end{cases}$$

Note, simple symm. RW are actually RW on Graphs.

$\alpha \sim .18$



RW on a circle.

$$\lim_{n\to\infty} P(S_n \bmod 4 = 0) = \frac{1}{4}$$

Think of rolling a dice, result of the roll at time n, $\epsilon_n \in \{1, 2, \dots, 6\}$.

$S_n = \sum_1^n \epsilon_i$, S_n is divisible by 4.

st. $P\underset{\sim}{1}' = \underset{\sim}{1}'$

$X_n = S_n \bmod 4;\ X_0 = 0$

$\{X_n : n \ge 0\}$ is a M.C. on $S = \{0, 1, 2, 3\}$.

Doubly stochastic matrix if col. sums are 1 for all col.

$\underset{\sim}{1}P = \underset{\sim}{1}$.

$$P = \begin{array}{c|cccc|} & 0 & 1 & 2 & 3 \\ \hline 0 & p_4 & p_1+p_5 & p_2+p_6 & p_3 \\ 1 & p_3 & p_4 & p_1+p_5 & p_2+p_6 \\ 2 & \vdots & & & \\ 3 & & & & \end{array}$$

Column sum of P is 1 for each column

Ex: Two state M.C., State sp.

$S = \{0, 1\}$.

$$P = \begin{array}{c|cc|} & 0 & 1 \\ \hline 0 & p_{00} & p_{01} \\ 1 & p_{10} & p_{11} \end{array}$$

$p_{00} + p_{01} = 1$

$p_{10} + p_{11} = 1$

Ex: Gambler's Ruin

Let X_n denote the fortune of A after n trials. $X_0 = a$, $S = \{0, 1, \cdots, N\}$, $N = a+b$.

$$P = \begin{array}{c|cccccc|} & 0 & 1 & 2 & 3 & \cdots & N \\ \hline 0 & 1 & & & & & \\ 1 & q & 0 & p & & & \\ 2 & & q & 0 & p & & \\ \vdots & & & \ddots & \ddots & \ddots & \\ & & & & q & 0 & p \\ N & & \cdots & - & \cdots & 0 & 1 \end{array}$$

$q + r + p$.

$$f(i,\epsilon) = \begin{cases} i & \text{if } i \in \{0,N\} \\ i+\epsilon & \text{o.w.} \end{cases}$$

Two st. states $\{0\}$ & $\{N\}$.

Gambler's Ruin prob against inf. rich opponent.

$$S = \{0, 1, - - - \}.$$

$$P = \begin{array}{c|cccccccc} & 0 & 1 & - & - & i-1 & i & i+1 & \\ \hline 0 & 1 & & & & & & & \\ 1 & q & 0 & p & & & & & \\ i & - & - & & & q & 0 & p & \end{array}$$

$\{0\}$ is a sp. states

Birth & death chain.

A M.C. on the state sp. $\{0, 1, \ldots, N\}$, (finite)
or $\{0, 1, \ldots \ldots\}$ (infinite) is called a birth & death
chain if $p_{ij} = 0$ if $|i-j| > 1$.

Finite st. space birth & death chain.

$$
P = \begin{array}{c|cccccccccc}
 & 0 & 1 & 2 & \cdots & i-1 & i & i+1 & \cdots & & N \\
\hline
0 & r_0 & p_0 & & & & & & & & \\
1 & q_1 & r_1 & p_1 & & & & & & & \\
2 & & q_2 & r_2 & p_2 & & & & & & \\
\vdots & & & & & & \vdots & & & & \\
i & & & & & q_i & r_i & p_i & & & \\
\vdots & & & & & & & & \ddots & & \\
N & & & & & & & & & q_N & r_N
\end{array}
$$

Set $r_0 + p_0$

1

$= q_N + r_N$

$= q_i + r_i + p_i = 1$

Inf. st. sp. birth & death chain

$$
P = \begin{array}{c|ccccccc|}
 & 0 & 1 & 2 & \cdots & i-1 & i & i \\
\hline
0 & r_0 & p_0 & & & & & \\
1 & q_1 & r_1 & p_1 & & & & \\
2 & & & & & & & \\
\vdots & & & & & & & \\
i & & & & & q_i & r_i & p_i \\
\vdots & & & & & & & \ddots
\end{array}
$$

Ex: Ehrenfest's model of diffusion of gases. A total N no. of particles are distributed in container A & B.

Let X_n be the no of particles in cont A. $N - X_n$ is the no of particles in B.

Dynamics :- At every time pt. pick a particle at random & change it's container.

$$P(X_{n+1} = i+1 \mid X_n = i) = (N-i)/N = 1 - i/N.$$

$$P(X_n = i-1 \mid X_n = i) = \frac{i}{N}.$$

Comments: In the finite st. Gambler's ruin problem:-

$$P = \begin{array}{c} \begin{array}{cccccc} 0 & 1 & \cdots & & & N \end{array} \\ \begin{pmatrix} q & p & & & & \\ q & 0 & p & & & \\ & \ddots & \ddots & \ddots & & \\ & & & & 1 & 0 \end{pmatrix} \end{array}$$

→ reflected with hard bdy at 0 and absorbing at N.